

PUMP ALARM, TRIP AND CRITICALITY MATRIX

Document ID: PUMP-ATM-001

Revision: 1.0

Asset Class: Centrifugal Pump

Purpose: Alarm Management, Operational Risk Assessment, Shutdown Logic and AI Copilot Severity Classification

1. Purpose

This document defines:

- Alarm Thresholds
- Critical Limits
- Equipment Trip Logic
- Escalation Procedures
- Risk Classification
- AI Copilot Severity Rules

The objective is to ensure abnormal conditions are detected and addressed before equipment damage, production loss or safety incidents occur.

2. Alarm Philosophy

The refinery follows a four-level alarm hierarchy.

Level 1

Operator Awareness

Purpose:

Inform operator of developing abnormal conditions.

Action:

Monitor trend.

No maintenance action required.

Level 2

Maintenance Planning

Purpose:

Indicates developing degradation.

Action:

Create maintenance notification.

Level 3

Immediate Intervention

Purpose:

Significant equipment degradation.

Action:

Reliability engineer review required.

Level 4

Emergency Condition

Purpose:

Potential imminent equipment failure.

Action:

Immediate intervention or controlled shutdown.

3. Bearing Temperature Alarm Matrix

Telemetry:

bearing_temp

Condition	Temperature
Normal	50°C – 75°C
Level 1	75°C – 80°C
Level 2	80°C – 85°C
Level 3	85°C – 95°C
Level 4	>95°C

Operational Interpretation

Level 1

Possible Early Wear

Action:

Trend monitoring

Level 2

Progressive Bearing Degradation

Action:

Schedule inspection

Level 3

Advanced Bearing Failure

Action:

Maintenance planning

Level 4

Potential Bearing Seizure

Action:

Immediate shutdown assessment

4. Motor Temperature Alarm Matrix

Telemetry:

motor_temp

Condition	Temperature
Normal	55°C – 85°C
Level 1	85°C – 90°C
Level 2	90°C – 95°C
Level 3	95°C – 105°C
Level 4	>105°C

Possible Causes

- Overload
 - Bearing Drag
 - Electrical Issues
 - Cooling Failure
-

5. Vibration Alarm Matrix

Telemetry:

vibration

Condition	Vibration
Normal	0 – 4 mm/s
Level 1	4 – 5 mm/s
Level 2	5 – 7 mm/s
Level 3	7 – 10 mm/s
Level 4	>10 mm/s

Failure Relationships

High Vibration + High Bearing Temperature

Diagnosis:

Bearing Failure

High Vibration + Low Suction Pressure

Diagnosis:

Cavitation

High Vibration + Seal Leakage

Diagnosis:

Alignment Issue

6. Flow Rate Alarm Matrix

Telemetry:

flow_rate

Condition	Flow Rate
Normal	250 – 350 m ³ /hr

Condition	Flow Rate
Level 1	220 – 250 m ³ /hr
Level 2	200 – 220 m ³ /hr
Level 3	180 – 200 m ³ /hr
Level 4	<180 m ³ /hr

Business Impact

Low Flow Rate Causes:

- Production loss
 - Reduced throughput
 - Process instability
-

7. Suction Pressure Alarm Matrix

Telemetry:

suction_pressure

Condition	Pressure
Normal	2 – 5 bar
Level 1	1.8 – 2.0 bar
Level 2	1.5 – 1.8 bar
Level 3	1.2 – 1.5 bar
Level 4	<1.2 bar

Cavitation Risk

Pressure below 2 bar significantly increases cavitation probability.

Pressure below 1.5 bar requires immediate investigation.

8. Discharge Pressure Alarm Matrix

Telemetry:

discharge_pressure

Condition	Pressure
Normal	8 – 15 bar
Level 1	7.5 – 8 bar
Level 2	7 – 7.5 bar
Level 3	6 – 7 bar
Level 4	<6 bar

Potential Causes

- Seal Leakage
- Internal Wear
- Impeller Damage

9. Motor Current Alarm Matrix

Telemetry:

motor_current

Condition	Current
Normal	80 – 120 A
Level 1	120 – 130 A
Level 2	130 – 140 A
Level 3	140 – 150 A
Level 4	>150 A

Interpretation

Increasing current often indicates increasing mechanical resistance.

10. Lubrication Level Alarm Matrix

Telemetry:

lubrication_level

Condition	Level
Normal	70% – 100%
Level 1	60% – 70%
Level 2	50% – 60%
Level 3	40% – 50%
Level 4	<40%

Reliability Impact

Low lubrication accelerates:

- Bearing Wear
 - Temperature Rise
 - Vibration Growth
-

11. Seal Leakage Alarm Matrix

Telemetry:

seal_leakage_rate

Condition	Leakage
Normal	0 – 0.1 L/hr
Level 1	0.1 – 0.2 L/hr

Condition	Leakage
Level 2	0.2 – 0.5 L/hr
Level 3	0.5 – 0.8 L/hr
Level 4	>0.8 L/hr

Consequences

- Product Loss
 - Environmental Release
 - Safety Hazards
-

12. Pump Health Severity Matrix

AI Copilot shall calculate severity using:

- Failure Probability
 - RUL
 - Telemetry Deviation
 - Failure Mode
-

Severity 1

Healthy

Failure Probability:

<30%

RUL:

180 Days

Action:

Monitor

Severity 2

Degrading

Failure Probability:

30–60%

RUL:

90–180 Days

Action:

Maintenance Notification

Severity 3

At Risk

Failure Probability:

60–80%

RUL:

30–90 Days

Action:

Maintenance Planning

Severity 4

Critical

Failure Probability:

80–95%

RUL:

10–30 Days

Action:

Reliability Review

Severity 5

Emergency

Failure Probability:

95%

RUL:

<10 Days

Action:

Immediate Intervention

13. Emergency Shutdown Decision Matrix

Shutdown Evaluation Required If:

Bearing Temperature >95°C

OR

Vibration >10 mm/s

OR

Motor Temperature >105°C

OR

Seal Leakage >1 L/hr

OR

Failure Probability >98%

AND

RUL <5 Days

14. AI Copilot Escalation Logic

If:

Failure Probability >80%

AND

Bearing Temperature Rising

Generate:

High Risk Alert

If:

Failure Probability >90%

AND

RUL <15 Days

Generate:

Critical Reliability Alert

If:

Failure Probability >95%

AND

RUL <10 Days

Generate:

Immediate Maintenance Recommendation

If:

Failure Probability >98%

AND

RUL <5 Days

Generate:

Emergency Response Recommendation

15. Summary

Alarm management should never rely on a single telemetry parameter.

The highest confidence decisions are achieved by combining:

- Telemetry Trends
- Failure Probability
- Failure Mode Classification
- Remaining Useful Life
- Operational Context

This document serves as the primary alarm interpretation reference for refinery operators, reliability engineers and the Industrial AI Copilot.